

Name: _____

SOLUTIONS

Date: _____

**Subject: Year 8 Mathematics****Test 4, 2015****Topics – Linear Relationships**

46
= _____ %

Total Time: 60 minutes
Total Reading: 5 minutes
Total Working: 55 minutes
Weighting: 15% of the year.
Equipment Allowed: 1 page notes (A4 one side)

CALCULATOR FREE**(6 marks, 1, 1, 1, 1, 1, 1)**

1. Complete the sentences from the list of words below;

Cartesian
x-axis**Linear**
Variable**Graph**
y intercept**Gradient****y- axis**

The Cartesian plane has two axes. The Horizontal axis is called the x-axis and the vertical axis is called the y-axis.

A rule that produces a straight line is called a Linear function.

The Gradient of a line is also called the slope and the y-intercept is where the line touches the y axis.

(4 marks , 2, 2)

2. Complete the tables below

a.

x	1	2	3	4	5	6
y	5	8	11	14	17	20

b.

x	1	2	3	4	5	6
y	13	8	3	-2	-7	-12

-1/2 Mark for errors

-1/2 Mark for errors

(4 marks, 2,2)

3. Write **an equation** for each of the following;

- a. nine is subtracted from a number and twenty one is the result.

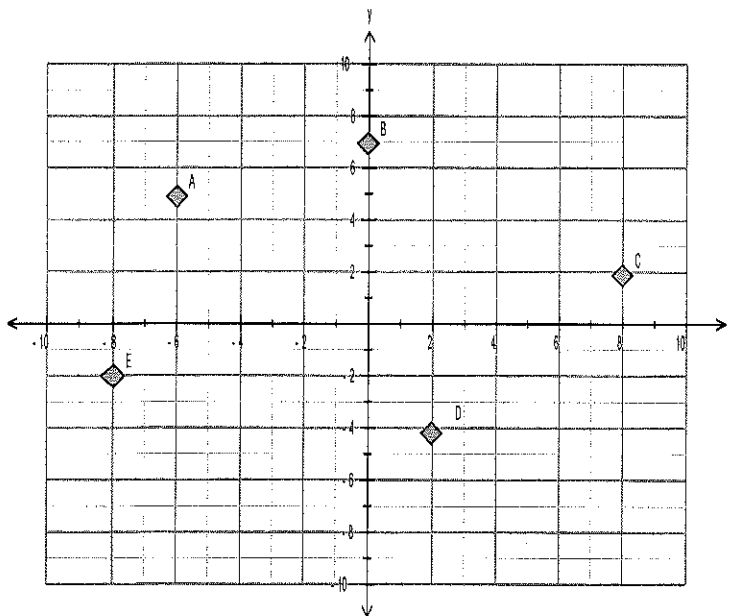
$$x - 9 = 21 \quad \checkmark \checkmark$$

- b. Three times a number, then add eight and one hundred and five is the result

$$3x + 8 = 105 \quad \checkmark \checkmark$$

(5 marks, 1,1,1,1,1)

4. Write down the coordinates for A, B, C, D and E.



A(-6, 5) ✓

B(0, 7) ✓

C(8, 2) ✓

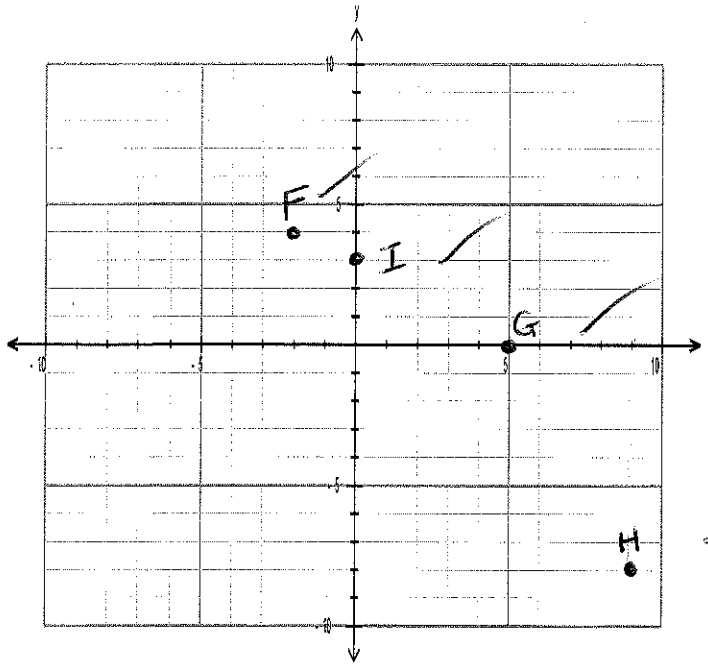
D(2, -4) ✓

E(-8, -2) ✓

(4 marks, 1,1,1,1)

5. Plot the following coordinates on the graph below.

F = (-2, 4), G = (5, 0), H = (9, -8), I = (0, 3)



(2 marks 1,1)

6. Solve the following equations.

a. $x + 3 = 17$

$x = 14$ ✓

b. $2x = 28$

$x = 14$ ✓

(4 marks 2, 2)

7. Complete the table below

a. $y = 4x + 2$

x	0	1	2	3	4	10
y	2	6	10	14	18	42

- 1/2 Mark
for errors.

✓✓

b. $y = -2x - 5$

x	0	1	2	3	4	10
y	-5	-7	-9	-11	-13	-25

- 1/2 Mark
for errors

✓✓

x	1	2	3	4	5	6
y	3	5	7	9	11	13

(3 marks 2, 1)

8.

a. From the table above write a rule in the form of $y = \dots$

✓
+2

✓
+2

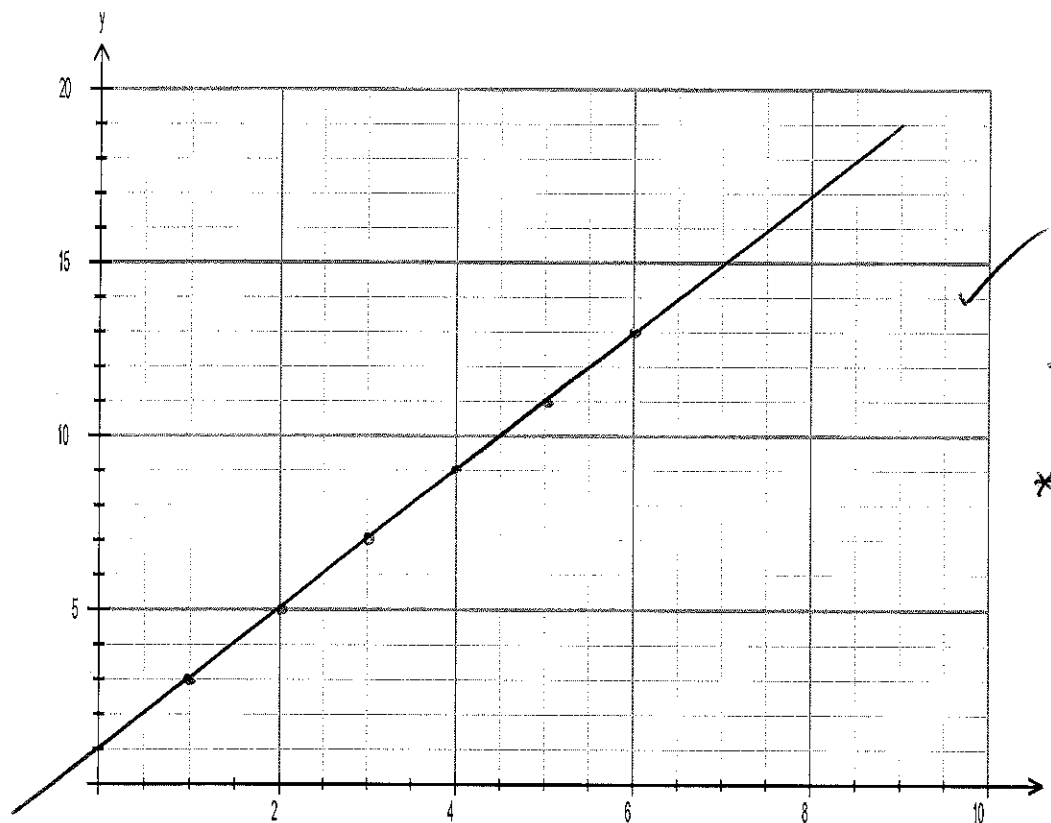
✓
+2

✓
+2

✓
+2

~~2x~~ + 1 ✓✓

b. Graph the table above.



- 1/2 mark.

* if no line is drawn just dots

CALCULATOR ALLOWED

Time: — 20 minutes	Marks for Section 2: 20
Reading: 2 minutes	Equipment Allowed: 1 A4 page of notes Calculator
Working: 22 minutes	

(2 marks, 1, 1)

9. Jayde is taking her family to watch a movie. She needs two adult tickets and 3 child tickets. Adult tickets cost \$21.50 and a child tickets cost \$12.00. This can be written as an equation

$$C = 21.50a + 12.00c$$

- a. How much would it cost Jayde to take her family to watch the film.

$$C = (21.50 \times 2) + (12.00 \times 3)$$

$$C = 43 + 36$$

$$C = \underline{\underline{\$79}} \quad \checkmark$$

- b. How much would it cost for 3 adults and 2 children?

$$C = (21.50 \times 3) + (12.00 \times 2)$$

$$C = 64.50 + 24$$

$$C = \underline{\underline{\$88.50}} \quad \checkmark$$

(3 marks 2, 1)

10. solve the following;

a. $4x + 8 = 24$

$$4x = 16 \quad \checkmark$$

$$x = \underline{\underline{4}} \quad \checkmark$$

b. $\frac{x}{7} = 13$

$$x = \frac{13}{7} \quad x = \underline{\underline{1.86}} \quad \checkmark$$

(4 marks, 1/2, 1/2, 2, 1)

11. Calculate the Gradient and the y intercept for the following.

a. $y = 5x - 2$

Gradient = 5

y intercept = -2

b. $2y + x = 4$

Gradient = $-\frac{1}{2}$

y intercept = 2

(3 marks, 2, 1)

12. A plumber charges \$25 an hour plus a set call out charge of \$15.

- a. create a rule to show the plumbers total cost (c), with hours (h)

$$C = 25h + 15$$

- b. How much would it cost to have the plumber for 12 hours

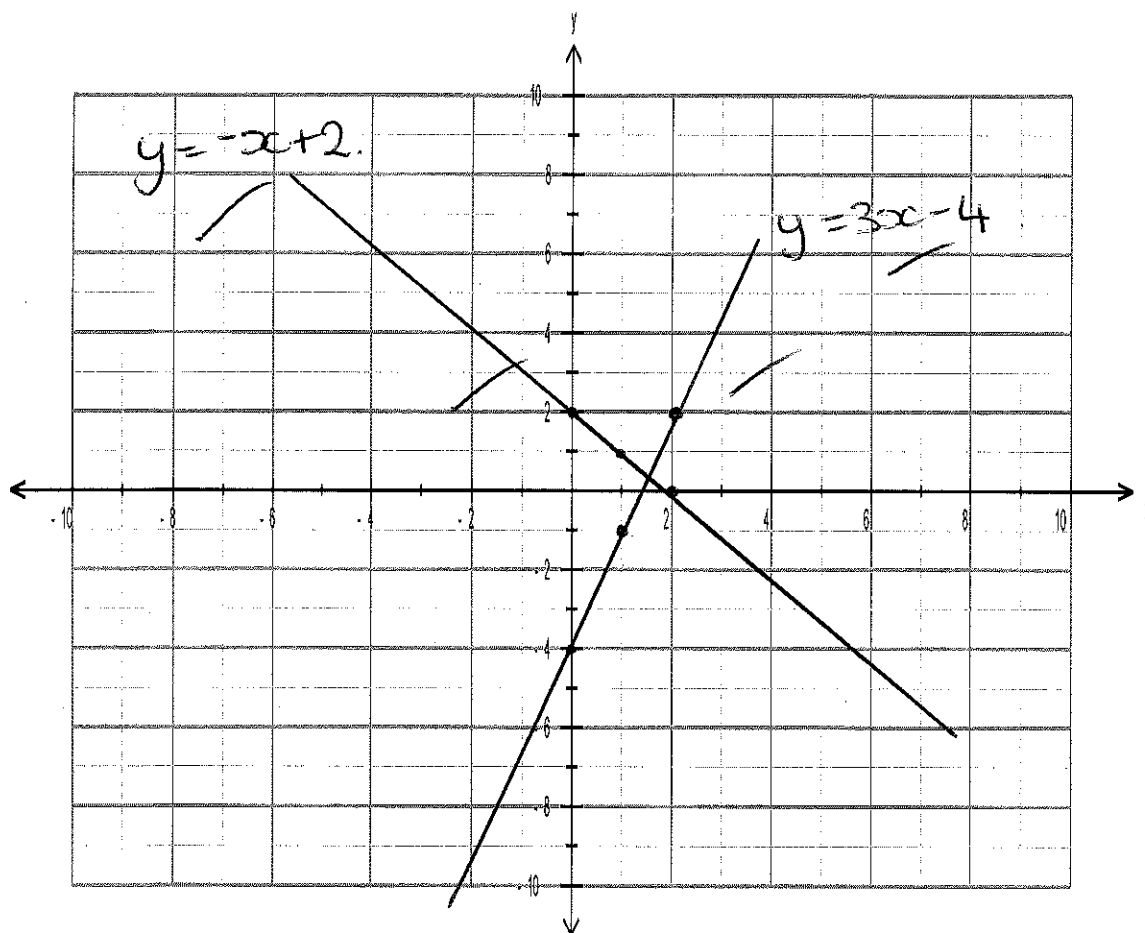
$$C = (25 \times 12) + 15 \quad C = \$315$$

(4 marks, 2, 2)

13. Draw the following equations on the Graph Below;

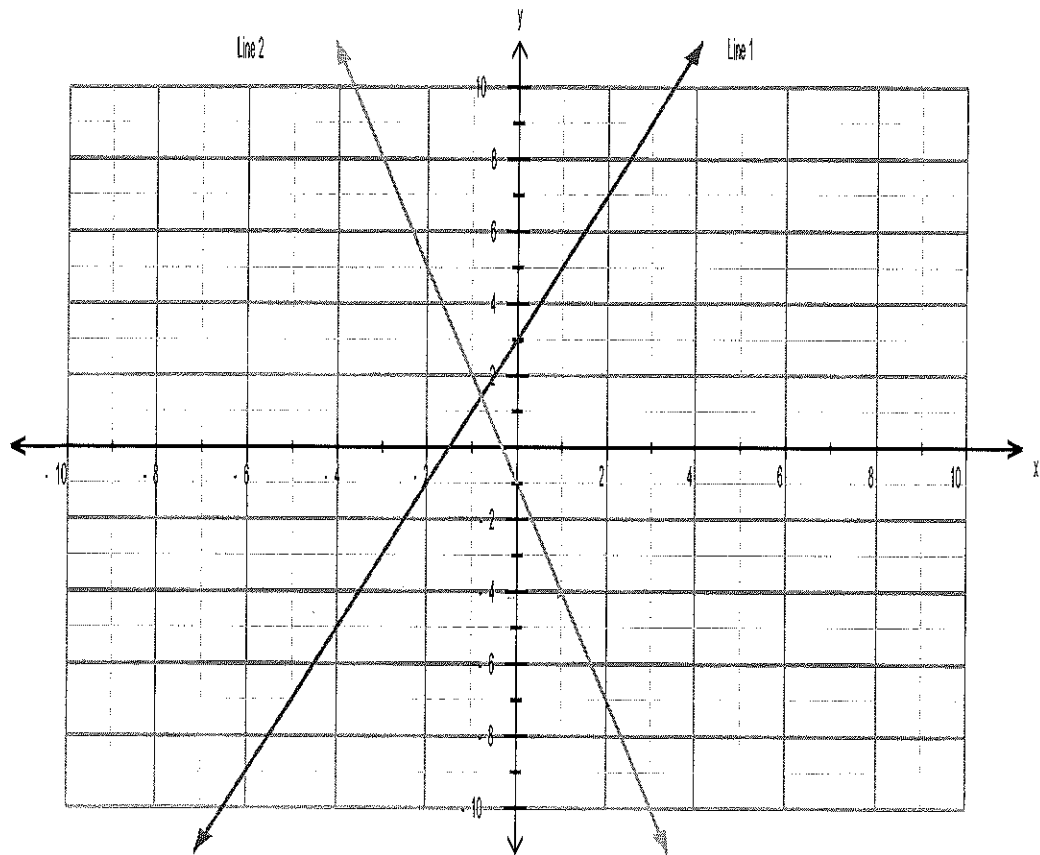
a. $y = -x + 2$

b. $y = 3x - 4$



(4 marks, 2, 2)

14. Find the rule for Line 1 and Line 2 from the graph below.



Line 1 $y = 2x + 3$ ✓ ✓

Line 2 $y = -3x + 3$ ✓ ✓